

Sunmook Choi, 최선묵

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EDUCATION

Ph.D. student in Applied Mathematics, Cornell University

Aug. 2024 – Present

- Advisor: [Sarah Dean](#)
- Committee members: David Bindel (Math), Robert Kleinberg (CS)
- Minors: Mathematics, Computer Science

M.S. in Mathematics, [Korea University](#)

Mar. 2022 – Aug. 2024

- Advisor: [Seungsang Oh](#)
- Thesis title: Three-dimensional dicube tilings of augmented Aztec bipyramids
- GPA: 4.5 / 4.5

B.S. in Mathematics, [Korea University](#)

Mar. 2016 – Feb. 2022

- Overall GPA: 4.3 / 4.5, Major GPA: 4.36 / 4.5
- Korean Mandatory Military Service (Auxiliary Police)

Jun. 2017 – Feb. 2019

RESEARCH INTERESTS

- **Machine Learning, Bandits, Reinforcement Learning, Control Theory**

PUBLICATIONS

Refereed Conference Publications

- [1] Y. Sattar, **S. Choi**, L. Maynard-Zhang, Y. Jedra, M. Fazel, S. Dean, *Two-Layer Linear Auto-Regressive Models Estimate Latent States*, to appear in ICML 2026
- [2] S. Chung, E. Kim, D. Kim, G. Heo, J. You, N. Lee, **S. Choi**, S. Han, S. Oh and I.-Y. Kwak, *BEAT2AASIST model with layer fusion for ESDD 2026 Challenge*, to appear in ICASSP 2026
- [3] **S. Choi**, Y. Sattar, Y. Jedra, M. Fazel and S. Dean, *Explore-then-Commit for Nonstationary Linear Bandits with Latent Dynamics*, to appear in AISTATS 2026
- [4] Y. Sattar, **S. Choi**, Y. Jedra, M. Fazel, S. Dean, *Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations*, the 64th IEEE Conference on Decision and Control (CDC), Rio (2025) 3862-3867
- [5] **S. Choi**, S. Chung, S. Lee, S. Han, T. Kang, J. Seo, I.-Y. Kwak and S. Oh, *TB-ResNet: Bridging the gap from TDNN to ResNet in Speaker Verification with Temporal-Bottleneck Enhancement*, 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul (2024) 10291-10295
- [6] **S. Choi**, I.-Y. Kwak and S. Oh, *Overlapped Frequency-Distributed Network: Frequency-Aware Voice Spoofing Countermeasure*, Proceedings of Interspeech 2022, Incheon, Korea (2022) 3558-3562
- [7] **S. Choi**, S. Oh, J. Yang, Y. Lee and I.-Y. Kwak, *Light-weight Frequency Information Aware Neural Network Architecture for Voice Spoofing Detection*, Proceedings of 2022 26th International Conference on Pattern Recognition (ICPR), Montreal (2022) 477-483

Workshop Publications

- [8] S. Han*, T. Kang*, **S. Choi***, J. Seo, S. Chung, S. Lee, S. Oh and I.-Y. Kwak, *CAU KU deep fake detection system for ADD 2023 challenge*, Proceedings of IJCAI 2023 Workshop on Deepfake Audio Detection and Analysis, Macao, S.A.R (2023) 23-30. (*: equal contribution)
- [9] I.-Y. Kwak, **S. Choi**, J. Yang, Y. Lee, S. Han and S. Oh, *Low-quality Fake Audio Detection through Frequency Feature Masking*, Proceedings of the 1st International Workshop on Deepfake Detection for Audio Multimedia (DDAM '22), New York (2022) 9-17

Journal Publications

- [10] S. Lee, **S. Choi**, T. Kang, S. Chung, S. Han, J. Seo, S. Park, E. Kim, S. Oh, I.-Y. Kwak, *iWAX: interpretable Wav2vec-AASIST-XGBoost framework for voice spoofing detection*, Scientific Reports **15** (1) (2025) 40491
- [11] S. Han, J. Seo, **S. Choi**, T. Kang, S. Chung, S. Lee, S. Park, S. Oh, I.-Y. Kwak, *Enhancing voice spoofing detection in noisy environments using frequency feature masking augmentation*, Engineering Science and Technology, an International Journal, **63** (2025) 101972
- [12] **S. Choi**, S. Lee and S. Oh, *Augmented Aztec bipyramid and dicube tilings*, Discrete Mathematics **347** (1) (2024) 113735

Preprints

- [13] D. Cao, B. Du, A. Lowitt, **S. Choi**, S. Dean, Y. Sattar, *Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control*, in submission
- [14] **S. Choi**, S. Lee and S. Oh, *Three-dimensional dicube tilings of an Aztec pyramid*, in submission
- [15] T. Kang, S. Han, **S. Choi**, J. Seo, S. Chung, S. Lee, S. Oh and I.-Y. Kwak, *Experimental Study: Enhancing Voice Spoofing Detection Models with wav2vec 2.0.*, [arXiv:2402.17127](https://arxiv.org/abs/2402.17127)

Thesis

- [16] *Three-dimensional dicube tilings of augmented Aztec bipyramids*, Master's thesis, Korea University [DOI link](#)

PRESENTATION **Oral Presentations**

- Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations, [64th IEEE Conference on Decision and Control \(CDC 2025\)](#), Rio De Janeiro, Brazil
- Aztec bipyramid and dicube tilings, [Knots and Spatial Graphs 2023](#), KAIST, Daejeon Korea.
- Aztec bipyramid and dicube tilings, [2022 Global KMS International Conference](#), Korean Mathematical Society (KMS), Seoul Korea.

Poster Presentations

- Explore-then-Commit for Nonstationary Linear Bandits with Latent Dynamics, [AISTATS 2026](#), Tanger, Morocco
- Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations, [North-east Systems and Control Workshop](#), Columbia University, USA
- Overlapped Frequency-Distributed Network: Frequency-Aware Voice Spoofing Countermeasure, [Interspeech 2022](#), Incheon, Korea
- Light-weight Frequency Information Aware Neural Network Architecture for Voice Spoofing Detection, [ICPR 2022](#), Montreal, Canada

Talks

- Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations, *Student Colloquium in Applied Mathematics (SCAM)*, Center for Applied Mathematics, Cornell University Dec. 2025
- My Know-how of TF Tutorial based on Problem Solving, *Tutorial TF Workshop* (online), Center for Teaching and Learning, Korea University Fall 2023, Spring 2024, Spring 2025

TEACHING EXPERIENCE

Teaching Assistant, Dept. of Mathematics, Cornell University

- Linear Algebra (MATH 4310) Spring 2026
- Linear Algebra for Engineers (MATH 2940) Fall 2025
- Linear Algebra for Engineers (MATH 2940) Spring 2025
- Linear Algebra for Engineers (MATH 2940) Fall 2024

Teaching Assistant, Dept. of Mathematics, Korea University

- Linear Algebra II With LAB (MATH 222) Fall 2023

	• Linear Algebra I With LAB (MATH 221) Spring 2023 • Linear Algebra II With LAB (MATH 222) Fall 2022 • Linear Algebra I With LAB (MATH 221) Spring 2022
AWARDS & HONORS	Fund Scholarship • KUAA 2024 Scholarship (Korea University Alumni Association in NY) Fall 2024 • Sigma (Korea Univ. Dept. of Mathematics Alumni Association) Fall 2021 • SAMSUNG Welfare Foundation Scholarship (Dream Class), SAMSUNG Spring 2019 Honors, Korea University • Dean's List, College of Science Fall 2020 • Semester High Honors Spring (2016, 2019, 2020, 2021), Fall (2019, 2020, 2021) Challenge Awards • ICASSP 2026 Grand Challenge on Environmental Sound Deepfake Detection Challenge (ESDD 2026) 3rd Dec. 2025 • IJCAI 2023 Competition and Challenge Award Aug. 2023 * Ranked in the 3rd place of Track 1.2 (Audio Fake Game - Detection, FG-D) from the Second Audio Deepfake Detection Challenge (ADD 2023) • ICASSP 2022 Grand Challenge Award Mar. 2022 * Ranked in the 3rd place of Track 1 (Low-quality Fake Audio Detection, LF) from the 1st Audio Deep Synthesis Detection Challenge (ADD 2022) Awards • Editors' Choice Award 2024, Discrete Mathematics Mar. 2024
SEMINARS	Deep Learning Seminar Season 1 & 2 (Leader) • https://github.com/smfelixchoi/MATH-DL-study-2 Jun. 2023 – Aug. 2023 • https://github.com/smfelixchoi/MATH-DL-study Sep. 2022 – Feb. 2023 • Theoretical Review and Python Implementation of SGD, MLP, CNN, RNN, AE, VAE, GAN. Deep Reinforcement Learning Seminar (Leader) Mar. 2023 – Sep. 2023 • https://github.com/smfelixchoi/MATH-DRL-study • Theoretical Review and Python Implementation of MDP, DP, RL, DQN, PG, AC, PPO.
WRITING SAMPLES	Summary Writing • Deep Reinforcement Learning (https://github.com/smfelixchoi/All-about-DRL, in progress) • Deep Learning (https://github.com/smfelixchoi/All-about-DL, in progress)
COMPETENCES	Languages Korean (native), English (proficient) Techniques $\text{\LaTeX}$, Python, TensorFlow, PyTorch, R, Julia
MENTORING	Directed Reading Program [related link] Spring 2026 • Mentoring an undergraduate student by reading a book or some papers Cornell Engineering High School Outreach Program Summer 2025 • CURIE Academy: Control & Navigation of Weather Balloons, led by Prof. Sarah Dean SAMSUNG Dream Class Spring 2019 • Teaching Mathematics to middle school students as an after-school program.

REFERENCES Available on request